PHYS 365 & 369 — Midterm Fall 2025 — Worked Solutions

25 questions. Every step uses the formula sheet printed at the back of this midterm (the same sheet as the current F2026 midterm formula sheet). Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	C	11	B	16	C	21	C
2	D	7	D	12	B	17	B	22	C
3	D	8	D	13	A	18	A	23	B
4	B	9	A	14	D	19	B	24	B
5	A	10	C	15	A	20	A	25	D

REMEMBER — not on the formula sheet

- Only the frequency is unchanged when a wave crosses a boundary (Q1, Q5).
- SHM velocity and acceleration: $v(t) = -\omega x_m \sin(\omega t + \phi)$, $v_{\max} = \omega x_m$, $a(t) = -\omega^2 x(t)$, $a_{\max} = \omega^2 x_m$ (Q13, Q21).
- Linear mass density of a cylindrical string: $\mu = \rho_{\text{material}}\, \pi \left(\frac{D}{2}\right)^2$, so $\mu \propto D^2$ (Q12).
- Molar mass of a gas mixture: $M_{\text{mix}} = \sum(\text{fraction}) \times M_i$ (Q8).
- Adding phasors: add the x and y components separately, then $A = \sqrt{A_x^2 + A_y^2}$ and $\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$, and check the quadrant (Q15).
- Consecutive resonances: the gap between them is $\Delta f = 2f_1$ for a closed–open pipe and $\Delta f = f_1$ for an open–open pipe (Q14).
- Average power on a string is proportional to the amplitude squared, $P \propto y_m^2$. This follows from the sheet's P_{avg} formula (Q23).

Question 1 Waves at a boundary

1. When a wave hits a boundary between two media, which wave property must be the same for all 3 of the incident, reflected, and transmitted waves?

A) The amplitude: $A_i = A_r = A_t$
B) The frequency: $\omega_i = \omega_r = \omega_t$
C) The wavenumber: $k_i = k_r = k_t$
D) The wave speed: $v_i = v_r = v_t$

Step 1. At a boundary the two media have to move together at the junction point, so the junction oscillates at one rate. That means the incident, reflected and transmitted waves all have the same frequency ω.

Step 2. The wave speed changes because $v = \sqrt{\dfrac{\tau}{\mu}}$ depends on the medium. Since $v = \dfrac{\omega}{k}$ with ω fixed, the wavenumber k changes too. The amplitudes change according to R and T.

$$\omega_i = \omega_r = \omega_t$$

Final answer: **(B) The frequency** ✓ Matches answer key

Question 2 Standing waves, string fixed at both ends

2. For a standing wave on a string fixed at both ends, the fifth normal mode of vibration frequency corresponds to:

A) One full wavelength fits into the string length.
B) One and a half wavelengths fit into the string length.
C) Two wavelengths fit into the string length.
D) Two and a half wavelengths fit into the string length.

FROM THE FORMULA SHEET

$f_n = \dfrac{v}{\lambda_n} = \dfrac{nv}{2L}$, for $n = 1, 2, 3, \ldots$

Step 1. From $\dfrac{v}{\lambda_n} = \dfrac{nv}{2L}$ we get $L = n\,\dfrac{\lambda_n}{2}$.

Step 2. The fifth mode has $n = 5$:

$$L = 5 \cdot \frac{\lambda}{2} = 2.5\,\lambda$$

So two and a half wavelengths fit on the string.

Final answer: **(D) Two and a half wavelengths** ✓ Matches answer key

Question 3 Power transmission coefficient

3. A string under tension T with linear mass density μ_1 is joined to a second string with linear mass density μ_2 that is 9 times larger than μ_1. A sinusoidal wave carrying power P is sent along string 1. What is the power transmission coefficient for the transmitted wave?



A) 1/3
B) 2/3
C) 1/4
D) 3/4

FROM THE FORMULA SHEET

$$Z = \sqrt{\mu\tau}, \quad T = \frac{2Z_1}{Z_1 + Z_2}, \quad T_e = \frac{Z_2}{Z_1} T^2$$

Step 1. Both strings are under the same tension and $\mu_2 = 9\mu_1$, so

$$\frac{Z_2}{Z_1} = \sqrt{\frac{\mu_2 \tau}{\mu_1 \tau}} = \sqrt{9} = 3 \quad \Rightarrow \quad Z_2 = 3Z_1$$

Step 2. Amplitude transmission coefficient:

$$T = \frac{2Z_1}{Z_1 + 3Z_1} = \frac{2}{4} = \frac{1}{2}$$

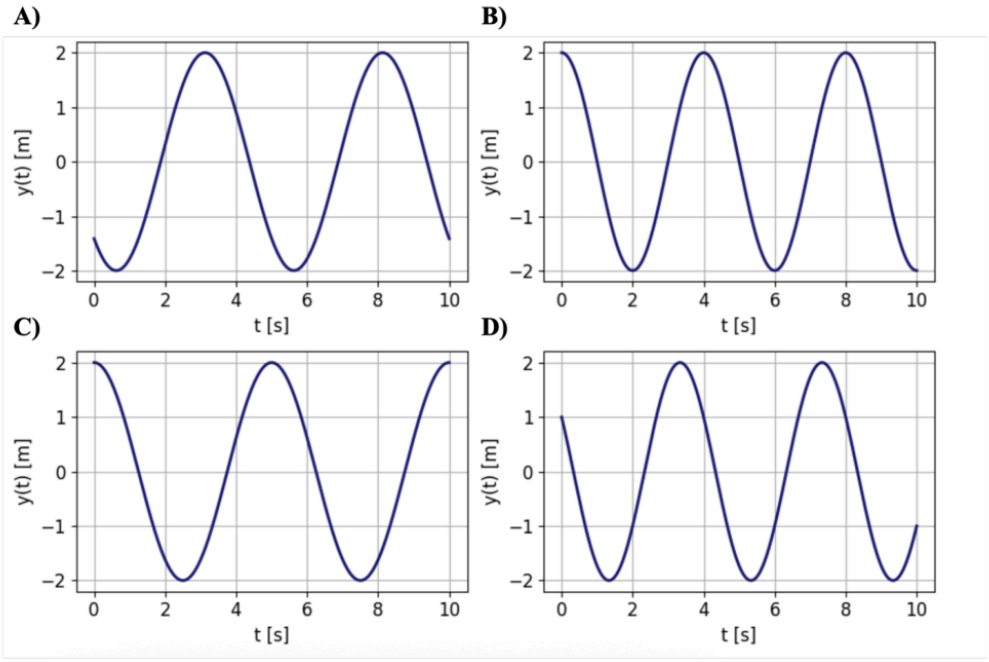
Step 3. Energy (power) transmission coefficient:

$$T_e = \frac{Z_2}{Z_1} T^2 = 3 \cdot \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

Final answer: **(D)** $\frac{3}{4}$ ✓ Matches answer key

Question 4 Reading an SHM graph

4. Which of the following is a correct graph of a simple harmonic oscillator with an amplitude of 2 m, an angular frequency of $\frac{\pi}{2}$, and a phase constant of 0?



FROM THE FORMULA SHEET

$$x(t) = x_m \cos(\omega t + \phi), \quad \omega = \frac{2\pi}{T}$$

Step 1. With $x_m = 2$ m, $\omega = \frac{\pi}{2}\ \frac{\text{rad}}{\text{s}}$ and $\phi = 0$: $y(t) = 2\cos\left(\frac{\pi}{2}t\right)$.

Step 2. Starting value: $y(0) = 2\cos(0) = +2$ m, so the graph must start at the top.

Step 3. Period:

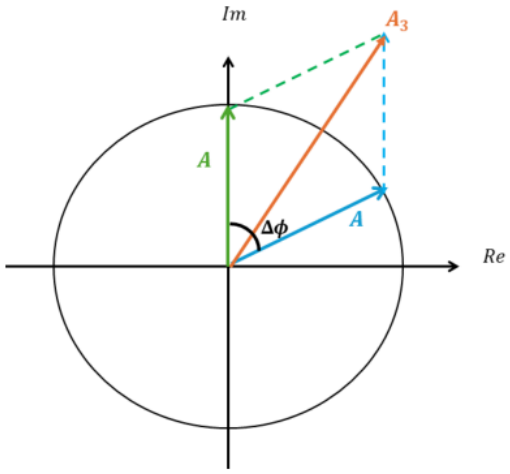
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{\pi}{2}} = 2\pi \cdot \frac{2}{\pi} = 4\ \text{s}$$

So the maxima are at $t = 0,\ 4,\ 8$ s and the minima at $t = 2,\ 6,\ 10$ s. Only graph B matches. Graph C starts at +2 but has $T = 5$ s.

Final answer: **(B) Graph B** ✓ **Matches answer key**

Question 5 Phasors crossing into a new medium

5. Two waves of equal amplitude A are traveling through the same medium with impedance Z_1. The picture shows the phasor representation of the two waves. Then the two waves cross a boundary into another medium with impedance $Z_2 > Z_1$. Which of the following statements regarding the phasor representation of the waves in the second medium is TRUE?



A) The angular frequency of the vectors spinning around the origin would stay the same, but the amplitudes of all the phasors would decrease.
B) The angular frequency of the vectors spinning around the origin and the amplitude of the resulting phasor would increase.
C) The amplitude of the initial phasors A and their phase difference $\Delta\phi$ would decrease such that the amplitude of the resulting phasor A_3 would stay the same.
D) The amplitudes and angular frequencies of all the phasors would remain unaffected by the new medium.

FROM THE FORMULA SHEET

$$T = \frac{2Z_1}{Z_1 + Z_2}$$

Step 1. The frequency does not change at a boundary, so the phasors keep spinning at the same angular frequency ω.

Step 2. Each wave's transmitted amplitude is $T \cdot A$. With $Z_2 > Z_1$:

$$T = \frac{2Z_1}{Z_1 + Z_2} < \frac{2Z_1}{Z_1 + Z_1} = 1$$

So both phasors shrink by the same factor T.

Step 3. Both waves are multiplied by the same T, so their phase difference $\Delta\phi$ stays the same. The resultant A_3 is just the whole diagram scaled down by T, so it shrinks as well.

Final answer: **(A) Same ω, all amplitudes decrease** ✓ **Matches answer key**

Question 6 Pipe length and harmonic frequency

6. A trombone has a variable length. When a musician blows into the mouthpiece and causes air in the tube of the horn to vibrate, the waves set up by the vibrations reflect back and forth in the horn to create standing waves. As the length of the horn is made shorter, what happens to the third harmonic frequency?

   A) The frequency remains the same.
   B) The frequency decreases.
   C) The frequency increases.
   D) The frequency increases or decreases depending on the diameter of the horn.

FROM THE FORMULA SHEET

$f_n = \dfrac{nv}{2L}$ (open–open) or $f_n = \dfrac{nv}{4L}$ (closed–open)

Step 1. In both formulas $f_n \propto \dfrac{1}{L}$, and v (air in the horn) and $n = 3$ stay fixed.

Step 2. Making L smaller makes f_3 larger. This is how a trombone plays higher notes when the slide is pulled in.

Final answer: **(C) The frequency increases** ✓ **Matches answer key**

Question 7 Complex amplitude

7. A simple harmonic oscillator oscillates according to $y(t) = 1.2\cos\left(2\pi t + \frac{\pi}{6}\right)$. What is the complex amplitude?

   A) $1.2\, e^{\frac{\pi}{6}}$
   B) $e^{j\frac{\pi}{6}}$
   C) $1.2e^{j\frac{\pi}{6}+j\omega t}$
   D) $1.2e^{j\frac{\pi}{6}}$

FROM THE FORMULA SHEET

$\tilde{A} = Ae^{j\theta} = A\left(\cos\theta + j\sin\theta\right)$

Step 1. Write the oscillation as the real part of a rotating phasor:

$$y(t) = 1.2\cos\left(2\pi t + \frac{\pi}{6}\right) = \mathrm{Re}\left[\,1.2\, e^{j\frac{\pi}{6}}\, e^{j\,2\pi t}\right]$$

Step 2. The complex amplitude is the time-independent part, which holds the amplitude and the phase constant:

$$\tilde{A} = 1.2\, e^{j\frac{\pi}{6}}$$

Option C still contains the time factor $e^{j\omega t}$, so it is not the complex amplitude.

Final answer: **(D)** $1.2\, e^{j\frac{\pi}{6}}$ ✓ Matches answer key

Question 8 Speed of sound in helium vs. air

8. What is the relationship between the speed of sound in atomic helium gas He compared to the speed of sound in air? Assume that both are at the same temperature T and that the air is made from 80% nitrogen N_2 and 20% oxygen O_2.
Mean atomic masses are: $M_{He} = 4.0026$ g/mol , $M_{N_2} = 28.014$ g/mol , $M_{O_2} = 32.00$ g/mol.

A) $v_{He} = 2.46\, v_{Air}$
B) $v_{He} = 0.37\, v_{Air}$
C) $v_{He} = 0.41\, v_{Air}$
D) $v_{He} = 2.93 v_{Air}$

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\gamma RT}{M}}, \quad \gamma_{\text{monoatomic}} = \frac{5}{3}, \quad \gamma_{\text{diatomic}} = \frac{7}{5}$$

REMEMBER — not on the formula sheet

Molar mass of a mixture: $M_{\text{air}} = 0.80\, M_{N_2} + 0.20\, M_{O_2}$

Step 1. Molar mass of the air in this problem:

$$M_{\text{air}} = 0.80(28.014) + 0.20(32.00) = 22.411 + 6.400 = 28.811 \ \tfrac{\text{g}}{\text{mol}}$$

Step 2. Helium is monatomic ($\gamma = \frac{5}{3}$). Air is N_2 and O_2, which are diatomic ($\gamma = \frac{7}{5}$). Take the ratio so that R and T cancel:

$$\frac{v_{He}}{v_{air}} = \sqrt{\frac{\gamma_{He}}{\gamma_{air}} \cdot \frac{M_{air}}{M_{He}}} = \sqrt{\frac{\frac{5}{3}}{\frac{7}{5}} \cdot \frac{28.811}{4.0026}}$$

Step 3. Evaluate. The unit $\frac{\text{g}}{\text{mol}}$ cancels in the ratio, so there is no need to convert to kg:

$$\frac{\gamma_{He}}{\gamma_{air}} = \frac{5}{3} \cdot \frac{5}{7} = \frac{25}{21} = 1.1905, \qquad \frac{M_{air}}{M_{He}} = 7.198$$

$$\frac{v_{He}}{v_{air}} = \sqrt{1.1905 \times 7.198} = \sqrt{8.569} = 2.93$$

Final answer: **(D)** $v_{He} = 2.93\, v_{air}$ ✓ Matches answer key

Question 9 Relativistic Doppler shift (light)

9. Hydrogen atoms in the laboratory emit red light with wavelength 656 nm. When the light from a distant galaxy is viewed, this spectral line of hydrogen is observed to have a wavelength of 691 nm. What is the speed of this galaxy relative to the earth?

A) 1.6×10^7 m/s
B) 6.4×10^7 m/s
C) 3.0×10^7 m/s
D) 7.6×10^7 m/s

FROM THE FORMULA SHEET

$f' = f\sqrt{\dfrac{c \pm v}{c \mp v}}, \quad c = f\lambda$ (from $v = f\lambda$)

Step 1. The observed wavelength is longer (691 nm compared with 656 nm), which is a red shift, so the galaxy is moving away. For a receding source use the lower signs:

$$f' = f\sqrt{\frac{c - v}{c + v}}$$

Step 2. Convert to wavelengths with $f = \dfrac{c}{\lambda}$:

$$\frac{c}{\lambda'} = \frac{c}{\lambda}\sqrt{\frac{c - v}{c + v}} \Rightarrow \left(\frac{\lambda'}{\lambda}\right)^2 = \frac{c + v}{c - v}$$

Step 3. Numbers: $\left(\dfrac{691}{656}\right)^2 = (1.05335)^2 = 1.10956 \equiv r$

Step 4. Solve for v: $r(c - v) = c + v \Rightarrow v = c\,\dfrac{r - 1}{r + 1}$

$$v = (3.00 \times 10^8)\,\frac{0.10956}{2.10956} = (3.00 \times 10^8)(0.05194) = 1.56 \times 10^7 \ \tfrac{\text{m}}{\text{s}}$$

Final answer: **(A)** $1.6 \times 10^7 \ \frac{\text{m}}{\text{s}}$ ✓ Matches answer key

Question 10 Mass–spring angular frequency

10. A spring ($k = 50$ N/m) oscillates on a frictionless horizontal surface with a 5.0 kg mass attached to it. What is the angular frequency of the oscillation?

A) 0.1 rad/s
B) 10 rad/s
C) 3.2 rad/s
D) 5 rad/s

FROM THE FORMULA SHEET

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{50}{5.0}} = \sqrt{10} = 3.16 \ \tfrac{\text{rad}}{\text{s}}$$

Final answer: **(C)** $3.2 \ \tfrac{\text{rad}}{\text{s}}$ ✓ **Matches answer key**

Question 11 Amplitude reflection coefficient

11. A wave on a string under tension F is incident from the left on a boundary where the linear mass density doubles. What is the amplitude reflection coefficient?



A) 0.17
B) -0.17
C) 0.33
D) -0.33

FROM THE FORMULA SHEET

$$Z = \sqrt{\mu\tau}, \quad R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

Step 1. Same tension F, so $Z \propto \sqrt{\mu}$. That gives $Z_1 = \sqrt{\mu F}$ and $Z_2 = \sqrt{2\mu F} = \sqrt{2}\, Z_1$.

Step 2. Substitute:

$$R = \frac{Z_1 - \sqrt{2}Z_1}{Z_1 + \sqrt{2}Z_1} = \frac{1 - \sqrt{2}}{1 + \sqrt{2}} = \frac{-0.4142}{2.4142} = -0.17$$

The minus sign means the reflected wave is inverted (reflection off a heavier string).

Final answer: **(B)** -0.17 ✓ **Matches answer key**

Question 12 Guitar string thickness

12. The low E and the high E strings on a guitar are the same length L, and are made from the same material. The high E string on a guitar vibrates with roughly 4 times the frequency of the low E string. Assuming both strings are under the same tension τ what is the relationship between the thicknesses/gauges of the two strings?
(Hint: Assume cylindrical strings. The wavelengths of the two waves are the same and are equal to twice the length of the string itself.)

A) The low E string is 2 times thicker than the high E string.
B) The low E string is 4 times thicker than the high E string.
C) The low E string is 8 times thicker than the high E string.
D) The low E string is 16 times thicker than the high E string.

FROM THE FORMULA SHEET

$$f = \frac{v}{\lambda}, \quad v = \sqrt{\frac{\tau}{\mu}}$$

REMEMBER — not on the formula sheet

For a cylindrical string of diameter D made of material with density ρ: $\mu = \frac{m}{L} = \rho\,\pi \left(\frac{D}{2}\right)^2 \Rightarrow \mu \propto D^2$

Step 1. Both strings have the same $\lambda = 2L$, so $f \propto v$. Then

$$\frac{v_{high}}{v_{low}} = \frac{f_{high}}{f_{low}} = 4$$

Step 2. Same tension, so $v \propto \frac{1}{\sqrt{\mu}}$:

$$\frac{v_{high}}{v_{low}} = \sqrt{\frac{\mu_{low}}{\mu_{high}}} = 4 \;\Rightarrow\; \frac{\mu_{low}}{\mu_{high}} = 16$$

Step 3. Same material, so $\mu \propto D^2$:

$$\frac{D_{low}^2}{D_{high}^2} = 16 \;\Rightarrow\; \frac{D_{low}}{D_{high}} = 4$$

Final answer: **(B) Low E is 4 times thicker** ✓ Matches answer key

Question 13 Max speed from position and acceleration

13. A mass of 1.5 kg is attached to a horizontal spring on a frictionless surface. At $t = 0$, the mass is released at +50 cm from equilibrium and accelerates at $-42\ \text{m/s}^2$. What is the maximum speed of the mass?

A) 4.6 m/s
B) 42 m/s
C) 0.68 m/s
D) 458 m/s

FROM THE FORMULA SHEET
$x(t) = x_m \cos(\omega t + \phi)$

REMEMBER — not on the formula sheet
Differentiate $x(t)$: $v = -\omega x_m \sin(\omega t + \phi)$ gives $v_{\max} = \omega x_m$. $a = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x$.

Step 1. The mass is released from rest at $x = +0.50$ m, so this is the amplitude: $x_m = 0.50$ m.

Step 2. Use $a = -\omega^2 x$ at that instant:

$$-42 = -\omega^2(0.50) \ \Rightarrow\ \omega^2 = \frac{42}{0.50} = 84 \ \Rightarrow\ \omega = 9.165\ \tfrac{\text{rad}}{\text{s}}$$

Step 3. Maximum speed:

$$v_{\max} = \omega x_m = (9.165)(0.50) = 4.58\ \tfrac{\text{m}}{\text{s}}$$

(The mass of 1.5 kg is not needed.)

Final answer: **(A)** $4.6\ \frac{\text{m}}{\text{s}}$ ✓ **Matches answer key**

Question 14 Identifying a pipe from its resonances

14. A pipe resonates in three successive normal modes at frequencies 431.25 Hz, 603.75 Hz and 776.25 Hz. The relevant speed of sound is 345 m/s. Based on this information, select the correct statement:

A) The pipe is open at both ends, and has length 1.0 m.
B) The pipe is open at one end, closed at the other, and has length 0.5 m.
C) The pipe is open at both ends, and has length 2.0 m.
D) The pipe is open at one end, closed at the other, and has length 1.0 m.

FROM THE FORMULA SHEET

Open–open: $f_n = \dfrac{nv}{2L},\ n = 1, 2, 3, \ldots$ Closed–open: $f_n = \dfrac{nv}{4L},\ n = 1, 3, 5, \ldots$

Step 1. Differences between successive modes: $603.75 - 431.25 = 172.5$ Hz and $776.25 - 603.75 = 172.5$ Hz.

Step 2. Test the open–open case, where the gap would be f_1, so $f_1 = 172.5$ Hz. Then $\dfrac{431.25}{172.5} = 2.5$, which is not a whole number, so the pipe is **not** open at both ends.

Step 3. Test the closed–open case. Only odd n occur, so successive modes differ by $2f_1$:

$$2f_1 = 172.5 \Rightarrow f_1 = 86.25 \text{ Hz}, \qquad \frac{431.25}{86.25} = 5, \quad \frac{603.75}{86.25} = 7, \quad \frac{776.25}{86.25} = 9$$

These are odd harmonics (5, 7, 9), which confirms a closed–open pipe.

Step 4. Length:

$$f_1 = \frac{v}{4L} \ \Rightarrow\ L = \frac{v}{4f_1} = \frac{345}{4(86.25)} = \frac{345}{345} = 1.0 \text{ m}$$

Final answer: **(D) Open at one end, closed at the other, $L = 1.0$ m** ✓ Matches answer key

Question 15 Adding AC currents as phasors

15. Two AC currents enter a 3-wire junction. The first is a current of 1.4 A at a phase of 180°, and the second is 2.5 A at 60°. What is the phase shift of the resultant current?

A) 94.0°
B) –86.0°
C) 176°
D) 209°

FROM THE FORMULA SHEET

$\tilde{A} = Ae^{j\theta} = A(\cos\theta + j\sin\theta)$

REMEMBER — not on the formula sheet

Phasor sum: add real parts and imaginary parts separately, then $\theta = \tan^{-1}\left(\dfrac{\text{Im}}{\text{Re}}\right)$. If Re < 0, add $180°$ to the calculator result.

Step 1. Write each current in component form:

$$\tilde{I}_1 = 1.4(\cos 180° + j\sin 180°) = -1.400 + 0j$$
$$\tilde{I}_2 = 2.5(\cos 60° + j\sin 60°) = 1.250 + 2.165j$$

Step 2. Add them: $\tilde{I} = (-1.400 + 1.250) + 2.165j = -0.150 + 2.165j$

Step 3. Angle: the real part is negative and the imaginary part is positive, so the result is in the 2nd quadrant.

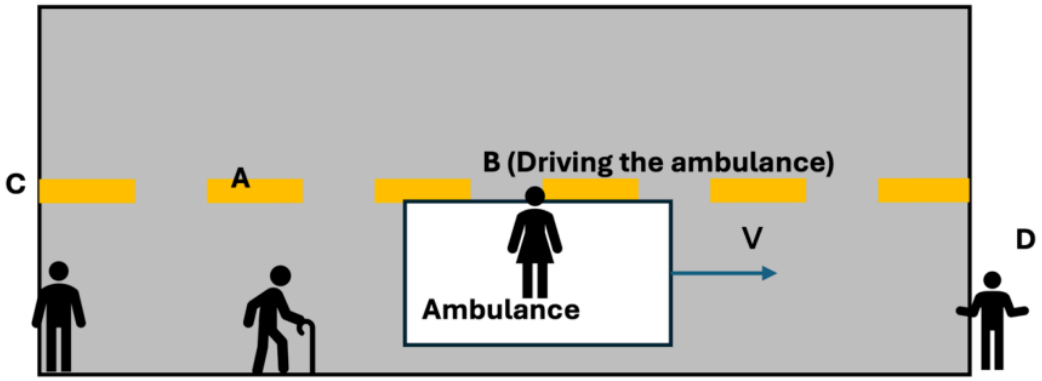
$$\tan^{-1}\left(\frac{2.165}{-0.150}\right) = -86.0^\circ \;\Rightarrow\; \theta = -86.0^\circ + 180^\circ = 94.0^\circ$$

(The calculator gives -86.0°, option B, which is the trap. That direction is in the 4th quadrant.)

Final answer: **(A)** 94.0° ✓ Matches answer key

Question 16 Doppler effect, ranking observers

16. An ambulance turns on its siren as it is driving with speed v. At that time, four observers are positioned as per the diagram below. A, C, and D are stationary, and B is driving the ambulance. Order the observers from highest measured frequency to lowest measured frequency.



A) $f_D > f_B = f_A = f_C$
B) $f_D > f_B = f_A > f_C$
C) $f_D > f_B > f_A = f_C$
D) $f_D > f_B > f_A > f_C$

FROM THE FORMULA SHEET

$f' = f\,\dfrac{v \pm v_D}{v \mp v_S}$ (top signs = toward)

Step 1. **D** is stationary and the source moves toward D: $f_D = f\,\dfrac{v}{v - v_S} > f$.

Step 2. **B** is in the ambulance and moves with the source, so there is no relative motion: $f_B = f$.

Step 3. **A** and **C** are both stationary and behind the ambulance, which is moving away from both: $f_A = f_C = f\,\dfrac{v}{v + v_S} < f$. How far away each one is does not matter. Only the velocities matter.

$$f_D > f_B > f_A = f_C$$

Final answer: **(C)** $f_D > f_B > f_A = f_C$ ✓ Matches answer key

Question 17 Superposition, finding the phase difference

17. A wave propagating on a string has the equation $y(x,t) = \frac{A}{2}\sin(kx - \omega t + \phi/2)$. If this wave is the result of the superposition of two initial waves with the same amplitude A, what is the phase difference ϕ between the two initial waves?

A) $60.33°$
B) $151.04°$
C) $0°$
D) $75.52°$

FROM THE FORMULA SHEET
$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right] \sin\left(kx - \omega t + \frac{1}{2}\phi\right)$

Step 1. Each initial wave has $y_m = A$. Match the resultant amplitude to $\frac{A}{2}$:

$$2A\cos\left(\frac{\phi}{2}\right) = \frac{A}{2} \;\Rightarrow\; \cos\left(\frac{\phi}{2}\right) = \frac{1}{4}$$

Step 2. Solve:

$$\frac{\phi}{2} = \cos^{-1}(0.25) = 75.52° \;\Rightarrow\; \phi = 151.04°$$

(Option D, $75.52°$, is $\frac{\phi}{2}$. Don't forget to double it.)

Final answer: **(B)** $151.04°$ ✓ **Matches answer key**

Question 18 Double slit with two wavelengths

18. Two monochromatic filters, one of which lets through light with wavelength $\lambda_1 = 650$ nm, are separately used in a Young's double slit experiment. The first (non-central) maximum of the interference pattern using filter 1 occurs in the same spot on the screen as the third maximum of the interference pattern using filter 2. What is the wavelength of light let through by the second filter?

A) $\lambda_2 = 217$ nm
B) $\lambda_2 = 1950$ nm
C) $\lambda_2 = 1300$ nm
D) $\lambda_2 = 325$ nm

FROM THE FORMULA SHEET
$d\sin\theta = m\lambda$ (bright fringes)

Step 1. Same spot on the screen means the same θ, so $d\sin\theta$ is the same for both:

$$m_1\lambda_1 = m_2\lambda_2 \;\Rightarrow\; (1)(650\text{ nm}) = (3)\lambda_2$$

Step 2. Solve:

$$\lambda_2 = \frac{650}{3} = 216.7\text{ nm} \approx 217\text{ nm}$$

Final answer: **(A)** $\lambda_2 = 217$ **nm** ✓ **Matches answer key**

Question 19 Clarinet (closed–open pipe)

19. The clarinet is well-modeled as a cylindrical pipe that is open at one end and closed at the other. For a clarinet whose air column has an effective length of 66 cm, determine the frequency f_3 of the third normal mode of vibration. Use 340 m/s for the speed of sound inside the instrument.

A) 193 Hz
B) 386 Hz
C) 579 Hz
D) 773 Hz

FROM THE FORMULA SHEET

$f_n = \dfrac{nv}{4L},\ n = 1, 3, 5, \ldots$

Step 1. The answer key takes "f_3" to mean the harmonic with $n = 3$, which is the third harmonic:

$$f_3 = \frac{3v}{4L} = \frac{3(340)}{4(0.66)} = \frac{1020}{2.64} = 386\text{ Hz}$$

NOTE ON THE ANSWER KEY

Strictly speaking, the "third normal mode" of a closed–open pipe is $n = 5$, which gives 644 Hz. That value is not an option. The exam uses the subscript $n = 3$ (the f_3 in the question), so the key's answer is 386 Hz. Watch for this wording on your exam.

Final answer: **(B) 386 Hz** ✓ **Matches answer key**

Question 20 Displacement amplitude from pressure amplitude

20. The pressure amplitude of a normal conversation sound wave is $\Delta p_m = 20$ mPa. The acoustic impedance of air is $Z_a = 412\ \text{Pa}\cdot\text{s/m}^3$. If we assume the frequency to be $f = 220$ Hz. What is the amplitude of the oscillation of the particles in the air?

A) 35 nm
B) 220 nm
C) 220 µm
D) 35 µm

FROM THE FORMULA SHEET
$\Delta p_m = (v\rho\omega)\, s_m, \quad Z = \rho v, \quad \omega = 2\pi f$

Step 1. Since $v\rho = Z$, the sheet formula becomes $\Delta p_m = Z\omega s_m$. Solve for s_m:

$$s_m = \frac{\Delta p_m}{Z\,\omega} = \frac{\Delta p_m}{Z\,(2\pi f)}$$

Step 2. Numbers, with 20 mPa $= 0.020$ Pa:

$$s_m = \frac{0.020}{(412)(2\pi)(220)} = \frac{0.020}{5.695\times 10^{5}} = 3.5\times 10^{-8}\ \text{m} = 35\ \text{nm}$$

Final answer: **(A) 35 nm** ✓ Matches answer key

Question 21 Changing ω at fixed k

21. A wave travels according to the equation $y(x,t) = A\cos(kx - \omega t + \phi_0)$. If the frequency is increased while the wavenumber remains the same, which of the following is true?

A) The wave speed will increase.
B) The maximum particle velocity will increase.
C) Both the wave speed and maximum particle velocity will increase.
D) Neither the wave speed nor maximum particle velocity will increase.

FROM THE FORMULA SHEET
$v = \dfrac{\omega}{k}$

REMEMBER — not on the formula sheet
Transverse particle velocity: $u = \dfrac{\partial y}{\partial t} = A\omega\sin(kx - \omega t + \phi_0) \;\Rightarrow\; u_{\max} = A\omega$

Step 1. Wave speed: $v = \dfrac{\omega}{k}$. If ω goes up while k stays fixed, v goes up.

Step 2. Maximum particle speed: $u_{\max} = A\omega$. If ω goes up, $u_{\max}$ goes up.

Final answer: **(C) Both increase** ✓ Matches answer key

Question 22 Beats, finding the second wave

22. A speaker emits a sound wave that can be represented by the equation $\Delta P_1 = P_{max\,1}\cos(8.04x - 880\pi t)$. A second speaker right beside the first emits a sound wave at a slightly different frequency, resulting in a beat frequency of 6 Hz. Assuming the second speaker is emitting a higher frequency than the first speaker, which of the following equations could be used to describe this sound wave?

A) $\Delta P_2 = P_{max\,2}\cos(8.04x - 892\pi t)$
B) $\Delta P_2 = P_{max\,2}\cos(8.09x - 886\pi t)$
C) $\Delta P_2 = P_{max\,2}\cos(8.15x - 892\pi t)$
D) $\Delta P_2 = P_{max\,2}\cos(8.04x - 886\pi t)$

FROM THE FORMULA SHEET

$f_{beat} = f_1 - f_2, \quad \omega = 2\pi f, \quad v = \dfrac{\omega}{k}$

Step 1. Speaker 1: $\omega_1 = 880\pi$ gives $f_1 = \dfrac{880\pi}{2\pi} = 440$ Hz, and $k_1 = 8.04\ \frac{\text{rad}}{\text{m}}$.

Step 2. Speaker 2 is higher by 6 Hz: $f_2 = 446$ Hz, so $\omega_2 = 2\pi(446) = 892\pi\ \frac{\text{rad}}{\text{s}}$.

Step 3. Both waves travel in the same air, so v is the same. A higher ω therefore needs a higher k:

$$v = \frac{\omega_1}{k_1} = \frac{\omega_2}{k_2} \;\Rightarrow\; k_2 = k_1\frac{\omega_2}{\omega_1} = 8.04 \cdot \frac{892}{880} = 8.15\ \frac{\text{rad}}{\text{m}}$$

Option A keeps $k = 8.04$, which would change the speed of sound, so it is wrong.

Final answer: **(C)** $\Delta P_2 = P_{max\,2}\cos(8.15x - 892\pi t)$ ✓ Matches answer key

Question 23 Energy carried vs. amplitude

23. I am shaking one end of a thick rope so there is a wave traveling along it. I am doing this exercise for 2 minutes. If I increase the amplitude of the wave by 22.5% and do the exercise again for 2 minutes, which is the TRUE statement?

A) The energy transported down the rope increases by 22.5%.
B) The energy transported down the rope increases by 50%.
C) The energy transported down the rope increases by 100%.
D) The energy transported down the rope increases by 122.5%.

FROM THE FORMULA SHEET

$$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$$

Step 1. Same rope (μ, v) and same shaking frequency (ω), so $P \propto y_m^2$. The time (2 min) is the same, so energy $E = Pt$ is also $\propto y_m^2$.

Step 2. The amplitude goes up by 22.5%, so $y_m' = 1.225\, y_m$:

$$\frac{E'}{E} = (1.225)^2 = 1.50 \;\Rightarrow\; \text{a } 50\% \text{ increase}$$

Final answer: **(B) Increases by 50%** ✓ Matches answer key

Question 24 Doppler, source and observer approaching

24. An ambulance with its siren on is approaching a car which is moving toward the ambulance. An observer in the car perceives the siren's frequency as:

A) lower than the actual frequency.
B) higher than the actual frequency.
C) the same as the actual frequency.
D) More information about the speed of the car and ambulance is needed.

FROM THE FORMULA SHEET

$$f' = f\,\frac{v \pm v_D}{v \mp v_S}$$

Step 1. The detector (car) moves toward the source, so use $+v_D$ in the numerator. The source (ambulance) moves toward the detector, so use $-v_S$ in the denominator:

$$f' = f\,\frac{v + v_D}{v - v_S}$$

Step 2. The numerator is $> v$ and the denominator is $< v$, so $f' > f$ whatever the actual speeds are. No more information is needed.

Final answer: **(B) Higher than the actual frequency** ✓ Matches answer key

Question 25 White light in Young's double slit

25. A coherent source of white light is used in a Young's double slit experiment. Select the TRUE statement.

A) Each maximum of the interference pattern will be a different color.
B) All maxima will be of white light.
C) Each maximum will be comprised of a rainbow of colors with the red end of the rainbow closest to the center.
D) The central maximum will be white, and each subsequent maximum will be comprised of a rainbow of colors with the blue end of the rainbow closest to the center.

FROM THE FORMULA SHEET

$d \sin\theta = m\lambda$

Step 1. Central maximum, $m = 0$: $\sin\theta = 0$ for every λ. All colours overlap at the centre, so it is white.

Step 2. For $m \geq 1$: $\sin\theta = \dfrac{m\lambda}{d}$, so a smaller λ gives a smaller θ. Blue (short λ) lands closer to the centre than red (long λ), so each side maximum is a little rainbow with blue on the inside.

Final answer: **(D) White centre, rainbows with blue closest to centre** ✓ Matches answer key